

Simran Nayak

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EXPERIENCE

Build Student Consultant, Data Science | The Build Fellowship | Feb 2025 – Present

Project: TweetLens – Unveiling Insights from Twitter Data

- Leveraging Twitter API and ChatGPT to programmatically collect real and synthetic tweet data, simulating user behavior, and enriching sentiment and engagement analysis
- Cleaning, preprocessing, and transforming unstructured text data into structured formats using data wrangling techniques for in-depth analysis
- Creating a Bag-of-Words WordCloud to visualize the most frequent terms, uncovering key customer sentiments, and trending topics
- Developing insightful visualizations using Python libraries such as Matplotlib and Seaborn to identify patterns, sentiment shifts, and topic clusters in Twitter activity

Project: Design an End-to-End Housing Price Prediction Machine Learning System

- Developed a robust ML system using Python to predict house prices based on 80+ features, incorporating EDA to identify key market drivers and pricing trends
- Engineered features and applied data preprocessing techniques to improve model accuracy and reduce noise in housing datasets
- Trained and fine-tuned supervised models (Linear Regression, XGBoost), achieving a 23% boost in prediction accuracy, and evaluated performance using RMSE, R², and MAE
- Deployed the final model on AWS SageMaker with a Gradio-powered dashboard, automating 75% of the pipeline for seamless, real-time predictions through an intuitive web interface

Software Quality Analyst | L Soft Corporation | September 2024 – January 2025

- Participated in SAP S/4 HANA data migration, handling data extraction, cleansing, transformation, and validation to ensure seamless integration of Customer Master Data from legacy systems
- Collaborated with the data load team to design accurate and efficient data load templates, enhancing data transfer quality and reliability
- Automated data pipelines using Azure SQL and Python, leading to a 60% reduction in manual effort and improved data consistency
- Conducted trend analysis on SAP Sales Order tables using Power BI and Excel, generating actionable insights for data-driven decision-making.

PROJECTS

Retail Sales Analysis & Forecasting

- Leveraged Python and Google BigQuery for scalable data preprocessing and querying, resulting in a 60% improvement in data quality and speed of retail trend analysis and customer segmentation
- Engineered predictive sales forecasting models using regression, ANOVA, and t-tests, significantly boosting inventory accuracy and enabling proactive stock management
- Developed dynamic Power BI dashboards integrated with BigQuery to deliver real-time sales insights, enhancing stakeholder decision-making and uncovering key performance drivers

Amazon Reviews Sentiment Analysis

- Performed sentiment analysis on Amazon reviews using Python, classifying feedback into positive, negative, and neutral categories
- Applied advanced text preprocessing techniques (cleaning, N-gram analysis, custom stopword removal), addressed class imbalance using SMOTE, and improved model accuracy
- Visualized frequent terms with a Bag-of-Words WordCloud to uncover key customer sentiments and emerging trends.

Boston Crime Analysis

- Conducted a comprehensive analysis of Boston's crime severity using R Studio, SQL, and Tableau, developing interactive dashboards to identify patterns and trends
- Applied statistical modeling to uncover factors contributing to crime rates and severity, collaborating with law enforcement agencies to present findings and provide actionable recommendations for crime prevention and resource allocation
- Implemented data validation and quality control measures, automating data extraction, preprocessing, and reporting workflows to ensure accuracy, reliability, and enhanced efficiency

EDUCATION

Master Of Science in Data Science & Analytics | Grand Valley State University • Grand Rapids, Michigan • 2024 • GPA: 3.7/4.0

SKILLS

Technical Skills: Python, SQL (MySQL), Machine Learning (Regressions, Hyper-parameter Tuning, XGBoost and Feature Selection), Sentiment and Text Analysis, R Programming (seaborns, matplotlib, plotly, ggplot2, shiny), MS Power BI, Tableau, SAS, MS Office, Gradio, AWS Sagemaker, Google BigQuery, Advanced Excel (Pivot Tables, VLOOKUP)

Analytical Skills: Data Mining, Predictive Modeling, Hypothesis Testing, Statistical Analysis, ANOVA, MANOVA, Regression Analysis, Exploratory Data Analysis